# Hierarchical Stability Notions and Lyapunov Functions for PDEs

Matthew M. Peet

arXiv:2603.17536v1 [math.OC] 18 Mar 2026

***Abstract*— Unlike linear ordinary differential equations (ODEs), linear partial differential equations (PDEs) admit a multitude of non-equivalent notions of stability. This variety makes interpretation of Lyapunov stability results challenging. To simplify this interpretation, we propose a framework for hierarchical classification of notions of stability and Lyapunov conditions. To do this, for every well-posed PDE and set of boundary conditions, we define a fundamental state on $L_2$ corresponding to the minimal information needed to uniquely forward propagate the solution. Stability notions and Lyapunov functions are then defined in terms of this fundamental state. This gives rise to a hierarchy of stability notions, the weakest being fundamental state to PDE state stability. Other stability notions and Lyapunov conditions may then be interpreted relative to this weakest notion. Hierarchies are established for: Lyapunov, exponential and finite-energy stability. Sufficient Lyapunov conditions are defined in terms of operator inequalities. Illustrative examples and computational tools are provided.**

## I. Introduction

Recent years have seen an increase in methods for stability analysis and control of partial differential equations (PDEs) without discretization. Often, these methods rely on Lyapunov functions to establish convergence of the system state, bound $L_2$ gain, or design observers and controllers. Unlike ODEs, for which we have a well-established universal state-space representation, PDEs take a variety of forms – with no clear and consistent notion of state space. As a result, basic definitions of stability vary from PDE to PDE and, even for a given PDE, will vary with the set of imposed boundary conditions[1]. The use of Lyapunov functions further complicates the problem. While it is accepted that a Lyapunov function should be positive, bounded, and non-increasing, the form taken by these upper and lower bounds is not consistent. As a result, even for a given PDE with a given set of boundary conditions, we may obtain a variety of non-equivalent stability results depending on how the Lyapunov function is defined and bounded.

To illustrate, consider a well-studied PDE: the wave equation $\mathbf{u}_{tt} = \mathbf{u}_{xx}$ with pinned boundary conditions $\mathbf{u}(t,0) = \mathbf{u}(t,1) = 0$. This PDE is not stable in the $L_2$-norm sense in that there exists no $C \geq 0$ such that $\|\mathbf{u}(t)\|_{L_2} + \|\dot{\mathbf{u}}(t)\|_{L_2} \leq C(\|\mathbf{u}(0)\|_{L_2} + \|\dot{\mathbf{u}}(0)\|_{L_2})$ for all $t \geq 0$. However, this PDE is stable in another sense, in that for $V(u) = \|\dot{u}\|_{L_2} + \|u_s\|_{L_2}$, $\dot{V} = 0$ and hence $\|\mathbf{u}_s(t)\|_{L_2} + \|\dot{\mathbf{u}}(t)\|_{L_2} \leq \|\mathbf{u}_s(0)\|_{L_2} + \|\dot{\mathbf{u}}(0)\|_{L_2}$. However, this tells us nothing about the actual PDE state, $\mathbf{u}(t)$. Fortunately, we know that $\|\mathbf{u}\| \leq C\|\mathbf{u}_s\|$ and hence $\|\mathbf{u}(t)\|_{L_2} + \|\dot{\mathbf{u}}(t)\|_{L_2} \leq C\|\mathbf{u}_s(0)\|_{L_2} + \|\dot{\mathbf{u}}(0)\|_{L_2}$ for some $C > 0$. However, as indicated, this bound cannot be strengthened to the first, more tradition notion of stability or well-posedness in $\mathbf{u}$. This illustration shows that obvious notions of stability often do not hold for PDEs (beam equations have the same problem). Nonetheless, it seems reasonable that we would want to be able to include wave equations in any Lyapunov framework for analysis and control of PDEs. Clearly, then, we must allow for some weaker notions of stability in the analysis and control of PDEs.

The need for a weaker notion of stability (and well-posedness) is well-known. Such notions for the wave equation can be found in [1] and other examples where stability depends on the choice of norm can be found in [2]. For a broader class of PDE, the framework developed in [3]–[5] for non-coercive Lyapunov functions weakens the positivity condition $V(\mathbf{u}) \geq \epsilon\|\mathbf{u}\|^2$ to $V(\mathbf{u}) > 0$ for all $\mathbf{u} \neq 0$. A strict negativity condition on the derivative of the Lyapunov function is then used to upper bound the solutions. This is similar the use of storage functions. However, the difficulty in proving stability of PDE's such as the wave equation lies not in positivity of the Lyapunov function, but in the upper bound. For example the Lyapunov function used for the wave equation discussed above is coercive in $\|\mathbf{u}\|$. However, it is not upper bounded by $\|\mathbf{u}\|$.

The contribution of this paper is relatively modest. We simply classify several properties of Lyapunov functions typically used in analysis of PDEs. These properties are divided into 3 categories: positivity (lower bounded), upper bounded, and negativity of the derivative. We then interpret those properties in terms of a hierarchy of notions of stability. The difficulty, as suggested above, is that the PDEs, boundary conditions, and Lyapunov functions vary substantially from case to case. To address this problem, the framework proposed in this paper is structured around the Partial Integral Equation (PIE) or fundamental state representation of PDEs. Specifically, if the boundary conditions of a PDE are suitably well-defined, there exists a bijection from $L_2$ to the set of sufficiently regular functions which satisfy the associated set of boundary conditions.

For example, consider the heat equation $\mathbf{u}_t = \mathbf{u}_{ss}$ where solutions are constrained to lie in some domain, say $X := \{\mathbf{u} \in H^2 \ : \ \mathbf{u}(0) = \mathbf{u}(1) = 0\}$. Then if we define $\mathcal{T}$ as

$$(\mathcal{T}\mathbf{x}) := \int_0^s \theta(s-1)\mathbf{x}(\theta)d\theta + \int_s^1 s(\theta-1)\mathbf{x}(\theta)d\theta$$

We have that $\mathcal{T} : L_2 \to X$, $\mathcal{T}\partial_s^2\mathbf{u} = \mathbf{u}$ and $\partial_s^2\mathcal{T}\mathbf{x} = \mathbf{x}$ for any $\mathbf{u} \in X$ and $\mathbf{x} \in L_2$. We refer to $\mathbf{x} := \mathbf{u}_{ss}$ as the fundamental state. Using this bijection, the dynamics of

M. Peet is with the School for the Engineering of Matter, Transport and Energy, Arizona State University. e-mail: mpeet@asu.edu

This work was supported by the National Science Foundation under grants No. 2429973 and 2337751.

[1]Semigroup theory provides basic definitions of state which apply to a large class of PDEs. However this approach simply shifts the problem into various definitions of inner product, domain, generator, et c.

the PDE may be equivalently [2] written as $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$. Of course, for the heat equation, this is simply[3] $\mathcal{T}\dot{\mathbf{x}} = \mathbf{x}$ and $\mathcal{T} = (\partial_s^2)^{-1}$ on the domain $X$. An equation of the form $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ where $\mathcal{T}$ and $\mathcal{A}$ have the structure given above is a Partial Integral Equations (PIEs). An operator $\mathcal{T}$ of this type is a Partial Integral (PI) operator and the set of PI operators define a linear algebra, with the set of PI operators with polynomial kernels being a sub-algebra. It has been shown that any PDE with suitably well-defined domain, $X$, admits such a representation with polynomial kernels [6]–[8].

Our approach, then, is to allow for classification of bounds on a Lyapunov function in terms of both the PDE state and the fundamental (PIE) state. Next, we classify notions of stability in terms of these bounds – focusing on two cases of Lyapunov stability: PDE stability ($\|\mathbf{u}(t)\| \leq C\|\mathbf{u}(0)\|$) and the weaker notion of PIE to PDE stability ($\|\mathbf{u}(t)\| \leq C\|\mathbf{x}(0)\|$). These notions are extended to exponential and finite-energy stability. This classification system then allows for interpretation of almost any PDE stability result which is based on the use of Lyapunov functions. Finally, to illustrate the application of these results, we express sufficient conditions for these various stability notions in terms of operator inequalities. These operator inequalities are applied to representative PDEs.

## II. Fundamental (PIE) State and Bijection $\mathcal{T}$

*Definition 1:* We say $\mathcal{P}$ is a PI operator, denoted $\mathcal{P} \in \Pi_3 \subset \mathcal{L}(L_2)$ if

$$(\mathcal{P}\mathbf{x}) := R_0(s)\mathbf{x}(s) + \int_0^s R_1(s,\theta)\mathbf{x}(\theta)d\theta + \int_s^1 R_1(s,\theta)\mathbf{x}(\theta)d\theta$$

where $R_0 \in L_\infty$ and $R_1, R_2 \in L_2$. Furthermore, we define $\Pi_2 := \{\mathcal{P} \in \Pi_3 \ : \ R_0 = 0\}$.

$\Pi_3$ and $\Pi_2$ are vector spaces and, when the $R_i$ are square, linear *-algebras under composition and $L_2$ adjoint. The subspaces of $\Pi_3$ and $\Pi_2$ with $R_i$ polynomial are likewise subalgebras. Furthermore, for $\mathcal{P} \in \Pi_3$ and $\mathcal{Q} \in \Pi_2$, we have $\mathcal{PQ}, \mathcal{QP} \in \Pi_2$. Extension to multivariate domains is the sum and composition of $\Pi_2, \Pi_3$ operators on each spatial dimension. Extension to mixed spatial domains such as $\mathcal{L}(\mathbb{R}^n \times L_2[0,1] \times L_2[[0,1]^2])$ can be found in [7], [8].

PDEs are typically defined in terms of $\dot{\mathbf{u}} = A\mathbf{u}$, where $A$ is a differential operator and $\mathbf{u}$ is constrained to lie in some subspace of a Sobolev space – which we refer to as $X$. If the PDE is well-posed, then $A : X \to L_2$ is invertible on $X$ and $\mathcal{T} := A^{-1} \in \Pi_3$. This results in an equation of the form $\mathcal{T}\dot{\mathbf{x}}(t) = \mathbf{x}(t)$ where we refer to $\mathbf{x}(t) \in L_2$ as the fundamental state. However, solving for $A^{-1}$ explicitly requires solution of certain differential equations. In [6], a framework was proposed whereby, roughly speaking, only the highest derivative, $D^\alpha$, in $A$ is inverted[4]. This results in a Partial Integral Equation (PIE) of the form $\mathcal{T}\dot{\mathbf{x}}(t) = \mathcal{A}\mathbf{x}(t)$ where $\mathcal{T} = (D^\alpha)^{-1} : L_2 \to X$ and $\mathcal{A} = A\mathcal{T} \in \Pi_3$. For this case, analytic formulae can be obtained for the polynomial kernels in $\mathcal{T}$. This is illustrated in the following lemma which concisely states the result in [6] and where we use $D^n = \partial_s^n$,

$$X = \Big\{ g \in H^n[a,b] \ : \quad \text{for } i = 1, \cdots, n \quad \sum_{j=1}^{n} \alpha_{i,j}(D^{j-1}g)(a) + \sum_{j=1}^{n} \beta_{i,j}(D^{j-1}g)(b) = 0 \Big\} \tag{1}$$

and

$$W_n(t) = \begin{bmatrix} 1 & t & \frac{1}{2}t^2 & \cdots & \frac{t^{n-1}}{(n-1)!} \\ 0 & 1 & t & \cdots & \frac{t^{n-2}}{(n-2)!} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & t \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

*Lemma 1:* For $X, \alpha_{i,j}, \beta_{i,j}$ as defined in Eq. (1), $[N_a]_{i,j} = \alpha_{i,j}$, $[N_b]_{i,j} = \beta_{i,j}$ if $det(N_a + N_b W(b-a)) \neq 0$, let $\mathcal{H} := D^n$ and

$$(\mathcal{T}\mathbf{x})(s) = \int_a^b G(s,\theta)\mathbf{x}(\theta)d\theta$$

$$G(s,\theta) = \begin{cases} \mathbf{e}_1(s-a)^T(I-P)\mathbf{e}_n(a-\theta) & \theta < s \\ -\mathbf{e}_1(s-a)^T P\mathbf{e}_n(a-\theta) & s < \theta \end{cases}$$

where $P = (N_a + N_b W(b-a))^{-1} N_b W(b-a)$ and

$$\mathbf{e}_1(s) = \begin{bmatrix} 1 \\ s \\ \frac{1}{2}s^2 \\ \vdots \\ \frac{s^{n-1}}{(n-1)!} \end{bmatrix}, \qquad \mathbf{e}_n(\theta) = \begin{bmatrix} \frac{\theta^{n-1}}{(n-1)!} \\ \vdots \\ \frac{1}{2}\theta^2 \\ \theta \\ 1 \end{bmatrix}.$$

Then for any $\mathbf{u} \in X$ and $\mathbf{x} \in L_2$, we have $\mathcal{H}\mathcal{T}\mathbf{x} = \mathbf{x}$, $\mathcal{T}\mathcal{H}\mathbf{u} = \mathbf{u}$ and $\mathcal{T}\mathbf{x} \in X$.

We refer to $\mathbf{u}$ as the PDE state and $\mathbf{x} = D^\alpha\mathbf{u}$ as the fundamental or PIE state. Stability notions and Lyapunov functions may then be defined in terms of the PIE state and, occasionally, interpreted in terms of the PDE state, $\mathbf{u} = \mathcal{T}\mathbf{x}$.

## III. Positivity, Boundedness, and Negativity

For each notion of stability which will be introduced in Sec. IV, we will associate a Lyapunov stability condition. Each of these Lyapunov conditions includes three parts: a lower bound on the function (positivity), an upper bound on the function, and a bound on the negativity of the derivative of the function. Before introducing notions of stability, we will characterize useful variations on these 3 properties. Specifically, we list possible properties of a Lyapunov function, $V$, in terms of PIE state, $\mathbf{x} = D^\alpha\mathbf{u}$ where the PDE state is obtained from the PIE state as $\mathbf{u} = \mathcal{T}\mathbf{x}$ for some PI operator, $\mathcal{T}$. This then implies that $\|\mathbf{u}\|_{L_2} \leq \|\mathcal{T}\|\|\mathbf{x}\|$. We start with positivity properties, from weakest to strongest.

*Definition 2 (Lyapunov positivity):* For a given operator $\mathcal{T} : L_2 \to X$ and Lyapunov function, $V : L_2 \to \mathbb{R}^+$ with $V(0) = 0$, we say that $V$ is

1) **Positive semidefinite** if $V(\mathbf{x}) \geq 0$ .
2) **PDE positive** if $V(\mathbf{x}) \geq \epsilon\|\mathcal{T}\mathbf{x}\|^2$ for some $\epsilon > 0$.
3) **PIE positive** if $V(\mathbf{x}) \geq \epsilon\|\mathbf{x}\|^2$ for some $\epsilon > 0$.

[2]For notational simplicity, we assume sufficient regularity for commutation of spatial and temporal derivatives. Otherwise we use $\partial_t(\mathcal{T}\mathbf{x}) = \mathcal{A}\mathbf{x}$.

[3]More generally, we may always write the dynamics of a PDE $\dot{u} = Au$ as $\mathcal{T}\dot{\mathbf{x}} = \mathbf{x}$ where $\mathcal{T} = A^{-1}$ on $X$. However, in this case, the kernels of operator $\mathcal{T}$ may not be polynomial and require analytic solution of certain differential equations. If such an operator *is* known, however, the results of this paper can be applied using this $\mathcal{T}$ and $\mathcal{A} = I$.

[4]If $A$ is not a differential operator, boundary conditions still require differentiation in order to obtain a bijection from $L_2$ to $X$.

These conditions are required to hold for all $\mathbf{x} \in L_2$.

*Note 1:* Note that we might have added that $V$ is **Positive definite** if $V(\mathbf{x}) > 0$ for all $x \neq 0$ – i.e. positivity in the non-coercive sense. However, this definition is somewhat difficult to interpret constructively. One possible mechanism is to say that $V$ is positive definite if $V(\mathbf{x}) \geq \epsilon \|\mathcal{R}\mathbf{x}\|^2$ for some invertible mapping $\mathcal{R}$ – so that PDE positivity then becomes a special case with $\mathcal{R} = \mathcal{T}$. This is similar in spirit to the non-coercive function proposed for the heat equation in [4] which can be expressed as $V(\mathbf{x}) = \langle \mathcal{T}\mathcal{T}\mathbf{x}, \mathcal{T}\mathcal{T}\mathbf{x} \rangle = \|\mathcal{T}\mathcal{T}\mathbf{x}\|^2$. The positivity types in Defn. 2 are hierarchical – PIE positive implies PDE positive implies positive definite implies positive semidefinite. The first holds because $V(\mathbf{x}) \geq \epsilon \|\mathbf{x}\|^2$ and $\|\mathcal{T}\mathbf{x}\| \leq \|\mathcal{T}\| \|\mathbf{x}\|$ imply $V(\mathbf{x}) \geq \epsilon \|\mathbf{x}\|^2 \geq \frac{\epsilon}{\|\mathcal{T}^2\|} \|\mathcal{T}\mathbf{x}\|^2$. However, PDE positivity does not imply PIE positivity, since there is no $C$ such that $\|\mathbf{x}\| \leq C \|\mathcal{T}\mathbf{x}\|$.

For upper bounds, from weak to strong, we have

*Definition 3 (Lyapunov bounds):* Given a Lyapunov function, $V : L_2 \to \mathbb{R}^+$ with $V(0) = 0$, we say that $V$ is

1) **PIE bounded** if $V(\mathbf{x}) \leq C \|\mathbf{x}\|^2$ for some $C > 0$.
2) **PDE bounded** if $V(\mathbf{x}) \leq C \|\mathcal{T}\mathbf{x}\|^2$ for some $C > 0$.

Here the inequalities must hold for and all $\mathbf{x} \in L_2$.

Any Lyapunov function of the form $V(\mathbf{x}) = \langle \mathbf{x}, \mathcal{P}\mathbf{x} \rangle$ with PI operator $\mathcal{P} \in \Pi_3$ is PIE bounded. Furthermore, PDE bounded implies PIE bounded since $\|\mathcal{T}\mathbf{x}\| \leq \|\mathcal{T}\| \|\mathbf{x}\|$.

For negativity, we have the following

*Definition 4 (Lyapunov Derivative Conditions):* Given a Lyapunov function, $V : L_2 \to \mathbb{R}^+$ with derivative $\dot{V}(\mathbf{x})$ where $\dot{V}(0) = 0$, we say that $\dot{V}$ is

1) **Negative semidefinite** if $\dot{V}(\mathbf{x}) \leq 0$
2) **PDE negative** if $\dot{V}(\mathbf{x}) \leq -\alpha \|\mathcal{T}\mathbf{x}\|^2$
3) **PIE negative** if $\dot{V}(\mathbf{x}) \leq -\alpha \|\mathbf{x}\|^2$
4) **Lyapunov Negative** if $\dot{V}(\mathbf{x}) \leq -\alpha V(\mathbf{x})$

for some $\alpha > 0$. Inequalities must hold for and all $\mathbf{x} \in L_2$.

Apart from "Lyapunov negative", negativity conditions are ordered from weakest to strongest – i.e. PIE negative implies PDE negative implies negative semidefinite. The first holds since $-\|\mathbf{x}\|^2 \leq -\frac{1}{\|\mathcal{T}\|^2} \|\mathcal{T}\mathbf{x}\|^2$. The strength of "Lyapunov negativity" relative to PIE and PDE negativity depends on positivity and boundedness of $V$. Specifically, Lyapunov negativity implies PDE or PIE negativity if $V$ is PDE or PIE positive, respectively. Likewise, PDE or PIE negativity implies Lyapunov negativity if $V$ is PDE or PIE, respectively.

## IV. Stability Notions and Lyapunov Functions

In this section, we define 4 notions of stability and, for each notion of stability, associate a sufficient condition in terms of existence of a Lyapunov function which satisfies a set of bounds as defined in Sec. III. For consistency, these notions of stability and Lyapunov conditions are all expressed in terms of the fundamental state, $\mathbf{x} \in L_2$. This formulation simplifies the presentation and the toolset used to verify the proposed Lyapunov conditions – since $\mathbf{x} \in L_2$ is not differentiable. When a condition involves the term $\mathcal{T}\mathbf{x} \in X$, this is interpreted as the state of the underlying PDE from whence the fundamental state was obtained and hence is restricted by boundary conditions and continuity constraints.

### A. Notions of Lyapunov Stability

Let us begin with the notions of Lyapunov stability. These notions are particularly important for energy-conserving systems such as wave and beam equations.

*Definition 5:* We say $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) **Lyapunov PIE to PDE stable** if $\|\mathcal{T}\mathbf{x}(t)\| \leq C \|\mathbf{x}(0)\|$
2) **Lyapunov PIE stable** if $\|\mathbf{x}(t)\| \leq C \|\mathbf{x}(0)\|$
3) **Lyapunov PDE stable** if $\|\mathcal{T}\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\|$
4) **Lyapunov PDE to PIE stable** if $\|\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\|$

for some $C > 0$ and any $\mathbf{x}(t)$ with $\mathcal{T}\dot{\mathbf{x}}(t) = \mathcal{A}\mathbf{x}(t)$.
While the notion of PDE to PIE stable is included for completeness, such a notion never holds when the PIE state is a partial derivative of the PDE state ($\mathbf{x} = D^\alpha \mathbf{u}$) since at time $t = 0$, this requires $\|\mathbf{x}(0)\| \leq C \|\mathcal{T}\mathbf{x}(0)\|$.In the following, we examine the relationships between the several proposed notions of Lyapunov stability.

*Lemma 2:* Given $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$, the following are true.

1) PIE stable implies PIE to PDE stable
2) PDE stable implies PIE to PDE stable
3) PDE to PIE stable implies PIE stable, PDE stable and PIE to PDE stable

*Proof:* For statement 1), we have $\|\mathbf{x}(t)\| \leq C \|\mathbf{x}(0)\|$ implies PIE to PDE stable since $\|\mathcal{T}\mathbf{x}(t)\| \leq \|\mathcal{T}\| \|\mathbf{x}(t)\| \leq \|\mathcal{T}\| C \|\mathbf{x}(0)\|$. For statement 2), we have $\|\mathcal{T}\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\|$ implies PIE to PDE stable since $\|\mathcal{T}\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\| \leq \|\mathcal{T}\| C \|\mathbf{x}(0)\|$. For statement 3), we have that $\|\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\|$ implies PDE stable since $\|\mathcal{T}(t)\| \leq \|\mathcal{T}\| \|\mathbf{x}(t)\| \leq \|\mathcal{T}\| C \|\mathcal{T}\mathbf{x}(0)\|$. Similarly, PIE stable follows since $\|\mathbf{x}(t)\| \leq C \|\mathcal{T}\mathbf{x}(0)\| \leq \|\mathcal{T}\| C \|\mathbf{x}(0)\|$. Statements 1) or 2) then imply PIE to PDE stable. ■

Lemma 2 shows that PIE to PDE stable is the weakest notion, since it is implied by any of the others. However, PDE stable does not imply PIE stable and PIE stable does not imply PDE stable.

*a) Lyapunov Conditions for Lyapunov Stability:* Having defined several notions of Lyapunov stability, we now provide conditions for such stability properties in terms of properties of a candidate Lyapunov function.

*Lemma 3 (Conditions for Lyapunov stability):* Suppose there exists $V : L_2 \to \mathbb{R}^+$ with $V(0) = 0$ and with $\dot{V}(\mathbf{x}) = \lim_{h \to 0^+} \frac{V(\hat{\mathbf{x}}(t+h)) - V(\mathbf{x})}{h}$ for any $\mathbf{x}(t)$ such that $\mathcal{T}\dot{\hat{\mathbf{x}}}(t) = \mathcal{A}\hat{\mathbf{x}}(t)$ with $\hat{\mathbf{x}}(0) = \mathbf{x}$. Then $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) PIE to PDE stable if $V$ is PDE positive, PIE bounded, and negative semidefinite.
2) PIE stable if $V$ is PIE positive, PIE bounded, and negative semidefinite.
3) PDE stable if $V$ is PDE positive, PDE bounded, and negative semidefinite.
4) PDE to PIE stable if $V$ is PIE positive, PDE bounded, and negative semidefinite.

*Proof:* In all cases, $\dot{V} \leq 0$ implies $V(\mathbf{x}(t)) \leq V(\mathbf{x}(0))$. Hence, for statement 1, we have

$$\epsilon \|\mathcal{T}\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) \leq C \|\mathbf{x}(0)\|^2.$$

For statement 2,

$$\epsilon \|\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) \leq C \|\mathbf{x}(0)\|^2.$$

For Statement 3,

$$\epsilon\|\mathcal{T}\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) \leq C\|\mathcal{T}\mathbf{x}(0)\|^2.$$

For statement 4,

$$\epsilon\|\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq V(\mathbf{x}(0)) \leq C\|\mathcal{T}\mathbf{x}(0)\|^2.$$

■

### B. Notions of Exponential Stability

For exponential stability, from weak to strong, we have

*Definition 6:* Given a well-posed PIE of the form $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$, we say the PIE is

1) **Exp. PIE to PDE stable** if $\|\mathcal{T}\mathbf{x}(t)\| \leq Ce^{-\alpha t}\|\mathbf{x}(0)\|$
2) **Exp. PIE stable** if $\|\mathbf{x}(t)\| \leq Ce^{-\alpha t}\|\mathbf{x}(0)\|$
3) **Exp. PDE stable** if $\|\mathcal{T}\mathbf{x}(t)\| \leq Ce^{-\alpha t}\|\mathcal{T}\mathbf{x}(0)\|$
4) **Exp. PDE to PIE stable** if $\|\mathbf{x}(t)\| \leq Ce^{-\alpha t}\|\mathcal{T}\mathbf{x}(0)\|$

for some $C, \alpha > 0$ and any $\mathbf{x}(t)$ with $\mathcal{T}\dot{\mathbf{x}}(t) = \mathcal{A}\mathbf{x}(t)$.

As in Lemma 2, Exp. PIE or PDE stable implies Exp. PIE to PDE stable and Exp. PDE to PIE stable implies all others.

**Lyapunov Conditions for Exponential Stability** To keep the conditions tight and concise, in Lemma 4 we use Lyapunov negativity instead of PIE or PDE negativity. This allows us to directly use $V(\mathbf{x}(t)) \leq -\alpha V(\mathbf{x}(t))$ to conclude $V(\mathbf{x}(t)) \leq e^{-\alpha t}V(\mathbf{x}(0))$. We will return to PIE and PDE negativity in Subsec. IV-C and Sec. V, while for now recalling only that the relation to Lyapunov negativity depends on PIE and PDE positivity and boundedness.

*Lemma 4 (Conditions for Exponential stability):* Suppose there exists $V : L_2 \to \mathbb{R}^+$ with $V(0) = 0$ and with $\dot{V}(\mathbf{x}) = \lim_{h\to 0^+} \frac{V(\hat{\mathbf{x}}(t+h))-V(\mathbf{x})}{h}$ for $\mathbf{x}(t)$ such that $\mathcal{T}\dot{\hat{\mathbf{x}}}(t) = \mathcal{A}\hat{\mathbf{x}}(t)$ with $\hat{\mathbf{x}}(0) = \mathbf{x}$. Suppose $\dot{V}$ is Lyapunov negative. Then $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) Exp. PIE to PDE stable if $V$ is PDE positive, PIE bounded.
2) Exp. PIE stable if $V$ is PIE positive, PIE bounded.
3) Exp. PDE stable if $V$ is PDE positive, PDE bounded.
4) Exp. PDE to PIE stable if $V$ is PIE positive, PDE bounded.

*Proof:* For statement 1, we have

$$\epsilon\|\mathcal{T}\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq e^{-\alpha t}V(\mathbf{x}(0)) \leq Ce^{-\alpha t}\|\mathbf{x}(0)\|^2.$$

For statement 2,

$$\epsilon\|\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq e^{-\alpha t}V(\mathbf{x}(0)) \leq Ce^{-\alpha t}\|\mathbf{x}(0)\|^2.$$

For statement 3,

$$\epsilon\|\mathcal{T}\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq e^{-\alpha t}V(\mathbf{x}(0)) \leq Ce^{-\alpha t}\|\mathcal{T}\mathbf{x}(0)\|^2.$$

For statement 4,

$$\epsilon\|\mathbf{x}(t)\|^2 \leq V(\mathbf{x}(t)) \leq e^{-\alpha t}V(\mathbf{x}(0)) \leq Ce^{-\alpha t}\|\mathcal{T}\mathbf{x}(0)\|^2.$$

■

### C. More Exotic Flavours: Finite Energy Stability

One might interpret "non-coercive" Lyapunov functions as $V$ positive definite, but not PDE positive. However, we don't have exactly a consistent mechanism to enforce $V$ positive definite and so we will relax this to $V$ positive semidefinite. In order to ensure a non-zero Lyapunov function, we then enforce a strict negativity condition on the derivative. A consequence of this formulation is that the Lyapunov function does not enforce a pointwise-in-time decay rate in the same way that exponential stability does, but rather only imposes a bound on the energy of the solution as measured by its *temporal* $L_2$ norm – i.e. $\|\mathbf{x}\|_{L_2} = \sqrt{\int_0^\infty \|\mathbf{x}(t)\|^2 dt}$. We refer to this notion of energy as "Finite-Energy" Stability.

*Definition 7:* Given a well-posed PIE of the form $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$, we say the PIE is

1) **PIE to PDE FE stable** if $\|\mathcal{T}\mathbf{x}\|_{L_2} \leq C\|\mathbf{x}(0)\|$
2) **PIE FE stable** if $\|\mathbf{x}\|_{L_2} \leq C\|\mathbf{x}(0)\|$
3) **PDE FE stable** if $\|\mathcal{T}\mathbf{x}\|_{L_2} \leq C\|\mathcal{T}\mathbf{x}(0)\|$
4) **PDE to PIE FE stable** if $\|\mathbf{x}\|_{L_2} \leq C\|\mathcal{T}\mathbf{x}(0)\|$

for some $C > 0$ and any $\mathbf{x}(t)$ which satisfies $\mathcal{T}\dot{\mathbf{x}}(t) = \mathcal{A}\mathbf{x}(t)$.

**Lyapunov Conditions for FE stability** Unlike exponential stability, negativity of the derivative is critical in FE stability. Hence, we do not use Lyapunov negativity in Lemma 5.

*Lemma 5 (Conditions for Finite Energy Stability):* Suppose $V$ is positive semidefinite. Then $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

- FE PIE to PDE stable if $V$ is PIE bounded and PDE negative.
- FE PIE stable if $V$ is PIE bounded and PIE negative.
- FE PDE stable if $V$ is PDE bounded and PDE negative.
- FE PDE to PIE stable if $V$ is PDE bounded and PIE negative.

*Proof:* In each case, we use $V(\mathbf{x}) \geq 0$ and integrate the Lyapunov negativity condition. Specifically, for the PDE negativity property, we integrate $\dot{V}(\mathbf{x}(t)) \leq -\alpha\|\mathcal{T}\mathbf{x}(t)\|^2$ to show that for any $T$, $V(\mathbf{x}(T)) \geq 0$ and hence

$$V(\mathbf{x}(T)) - V(\mathbf{x}(0)) \leq -\alpha \int_0^T \|\mathcal{T}\mathbf{x}(t)\|^2 dt$$
$$\Rightarrow \qquad \int_0^T \|\mathcal{T}\mathbf{x}(t)\|^2 dt \leq \frac{1}{\alpha}V(\mathbf{x}(0)).$$

We may then take the limit as $T \to \infty$ to show that

$$\alpha\|\mathcal{T}\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0)).$$

For PIE negativity, we similarly have $\alpha\|\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0))$. Now, statement 1) follows since $\alpha\|\mathcal{T}\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0)) \leq C\|\mathbf{x}(0)\|^2$.

Statement 2) from $\alpha\|\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0)) \leq C\|\mathbf{x}(0)\|^2$.

Statement 3) from $\alpha\|\mathcal{T}\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0)) \leq C\|\mathcal{T}\mathbf{x}(0)\|^2$.

Statement 4) from $\alpha\|\mathbf{x}\|_{L_2}^2 \leq V(\mathbf{x}(0)) \leq C\|\mathcal{T}\mathbf{x}(0)\|^2$. ■

*Note 2:* There is a significant question of whether FE stability implies asymptotic stability. Suppose we have FE PIE stable or FE PDE to PIE stable. Then $\mathbf{x} \in L_2$ and hence

$$\|\mathcal{T}\dot{\mathbf{x}}\|_{L_2} = \|\mathcal{A}\mathbf{x}\|_{L_2} \leq \|\mathcal{A}\|\|\mathbf{x}\|_{L_2}$$

Hence $\mathcal{T}\dot{\mathbf{x}} \in L_2$, which implies $\lim_{t\to\infty}\|\mathcal{T}\mathbf{x}(t)\| \to 0$. Unfortunately, however, this only holds for FE PIE stable and FE PDE to PIE stable.

## V. Sufficient Conditions for Notions of Stability

In this section, we briefly provide a list of relatively obvious conditions for the various stability notions in terms of operator inequality constraints on the PI operators $\mathcal{T}$ and $\mathcal{A}$, as well as some operator variable, $\mathcal{P} \in \Pi_3$ or $\mathcal{Q} \in \Pi_3$. Here the variable label $\mathcal{P}$ will be used when the variable is constrained to be self-adjoint and $\mathcal{Q}$ otherwise. The operator inequality constraints are all interpreted as positivity on $L_2$ and may be tested using discretization or analytically by finding a Mercer representation of the kernel. Alternatively, operator inequalities may be enforced numerically

using semidefinite programming through software such as PIETOOLS [9].

*Lemma 6 (LPIs for Lyapunov Bounds):* Given $\mathcal{T}$, if $V(\mathbf{x}) = \langle \mathbf{x}, \mathcal{P}\mathbf{x}\rangle$, then

1) positive semidefinite is equivalent to $\mathcal{P} \geq 0$
2) PDE positivity is equivalent to $\mathcal{P} \geq \epsilon \mathcal{T}^*\mathcal{T}$
3) PIE positivity is equivalent to $\mathcal{P} \geq \epsilon I$
4) PDE bounded is equivalent to $\mathcal{P} \leq C\mathcal{T}^*\mathcal{T}$
5) PIE bounded is equivalent to $\mathcal{P} \leq CI$

for some $\epsilon, C > 0$. If $\dot{V}(\mathbf{x}) = \langle \mathbf{x}, \mathcal{D}\mathbf{x}\rangle$, then

1) PDE negative is equivalent to $\mathcal{D} \leq -\alpha \mathcal{T}^*\mathcal{T}$
2) PIE negative is equivalent to $\mathcal{D} \leq -\alpha I$
3) Lyapunov negative is equivalent to $\mathcal{D} \leq -\alpha P$

for some $\alpha > 0$.

Note that PIE bounded in satisfied for any $\mathcal{P} \in \Pi_3$.

Every sufficient condition proposed in this section relies on one of the following two canonical candidate Lyapunov functions. Specifically, given $\mathcal{T}$, we have

$$V_1(\mathbf{x}) := \langle \mathcal{T}\mathbf{x}, \mathcal{P}\mathcal{T}\mathbf{x}\rangle \quad \text{and} \quad V_2(\mathbf{x}) := \langle \mathbf{x}, \mathcal{Q}\mathcal{T}\mathbf{x}\rangle$$

where $V_1$ and $V_2$ are parameterized by PI operators $\mathcal{P} \in \Pi_3$ and $\mathcal{Q} \in \Pi_3$, respectively. Here we constrain $\mathcal{P} = \mathcal{P}^*$ and $\mathcal{Q}\mathcal{T} = \mathcal{T}^*\mathcal{Q}^*$.

**Properties of $V_1$:** First consider $V_1(\mathbf{x})$, which is perhaps the most obvious form of Lyapunov function, being defined solely in terms of the PDE state, $\mathcal{T}\mathbf{x}$. Applying Lem. 6 with $\mathcal{P} \mapsto \mathcal{T}^*\mathcal{P}\mathcal{T}$, $V_1$ is PDE positive if $\mathcal{P} \geq \epsilon I$. However, it is not usually possible for $V_1$ to be PIE positive, since this would imply $\mathcal{T}^*\mathcal{P}\mathcal{T} \geq \epsilon I$. If we recall that typically $\mathcal{T} \in \Pi_2$ (or at least some part thereof), this implies that $\mathcal{T}^*\mathcal{P}\mathcal{T} \in \Pi_2$ and hence cannot be coercive. Next, we note that $V_1$ is both PDE and PIE bounded, since $\mathcal{T}^*\mathcal{P}\mathcal{T} \leq \|\mathcal{P}\|\mathcal{T}^*\mathcal{T}$ implies PDE bounded and PDE bounded implies PIE bounded. For $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$, the derivative of $V_1$ is

$$\dot{V}_1(\mathbf{x}) := \langle \mathcal{T}\mathbf{x}, \mathcal{P}\mathcal{A}\mathbf{x}\rangle + \langle \mathcal{P}\mathcal{A}\mathbf{x}, \mathcal{T}\mathbf{x}\rangle = \langle \mathbf{x}, (\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T})\mathbf{x}\rangle.$$

Thus $\dot{V}_1$ is PDE negative if $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq -\epsilon\mathcal{T}^*\mathcal{T}$. However, as for PIE positivity, $\dot{V}$ is unlikely to be PIE negative since this would require $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq -\epsilon I$ which is not possible unless $\mathcal{T}, \mathcal{A} \in \Pi_3$.

**Properties of $V_2$:** Consider now $V_2 = \langle \mathbf{x}, \mathcal{Q}\mathcal{T}\mathbf{x}\rangle$ where $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^*$. The difference between $V_1$ and $V_2$ is that $V_2$ is defined in terms of both the PDE state ($\mathcal{T}\mathbf{x}$) and PIE state ($\mathbf{x}$). For example, in the heat equation, for $\mathcal{Q} = -I$ we would have $V = -\langle \mathbf{u}, \mathbf{u}_{ss}\rangle$ which reduces to something like $V = \langle \mathbf{u}_s, \mathbf{u}_s\rangle$, depending on boundary conditions. Applying Lem. 6 with $\mathcal{P} \mapsto \mathcal{Q}\mathcal{T}$, $V_2$ is PDE positive if $\mathcal{Q}\mathcal{T} \geq \epsilon\mathcal{T}^*\mathcal{T}$. As with $V_1$, PIE positivity is not possible since it implies that $\mathcal{Q}\mathcal{T} \notin \Pi_2$. $V_2$ is always PIE bounded and is PDE bounded if $\mathcal{Q}\mathcal{T} \leq C\mathcal{T}^*\mathcal{T}$. For $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$, the derivative of $V_2$ is

$$\dot{V}_2(\mathbf{x}) := \langle \mathbf{x}, \mathcal{Q}\mathcal{A}\mathbf{x}\rangle + \langle \mathcal{Q}\mathcal{A}\mathbf{x}, \mathbf{x}\rangle = \langle \mathbf{x}, (\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^*)\mathbf{x}\rangle.$$

Thus $\dot{V}_2$ is PDE negative if $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq -\epsilon\mathcal{T}^*\mathcal{T}$ and PIE negative if $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq -\alpha I$. Unlike $\dot{V}_1$, it is possible for $\dot{V}_2$ to be PIE negative. Finally, note that $V_1$ is a special case of $V_2$ with $\mathcal{Q} = \mathcal{T}^*\mathcal{P}$.

### A. Operator Inequalities for Lyapunov Stability

In the following corollary, we combine $V_1$ and $V_2$ with Lemmas 3 and 6 to obtain the following operator inequalities for each category of Lyapunov stability.

*Corollary 7:* Given $\mathcal{T}, \mathcal{A}$, the PIE $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) PIE to PDE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon\mathcal{T}^*\mathcal{T}$ and $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq 0$.
2) PIE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon I$ and $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq 0$.
3) PDE stable if $\mathcal{P} \geq \epsilon I$, $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq 0$.
4) PDE to PIE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon I$, $\mathcal{Q}\mathcal{T} \leq C\mathcal{T}^*\mathcal{T}$, and $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq 0$.

for some $\mathcal{Q}$ or $\mathcal{P}$, $\epsilon, C > 0$.

*Example 1 (PIE to PDE Stability of the Wave Equation):* Let us now reconsider the wave equation which we recall is not PDE stable. Specifically, the wave equation $u_{tt} = u_{xx}$ with $u(t,0) = u(t,1) = 0$ admits a PIE representation of the form

$$\overbrace{\begin{bmatrix} I & 0 \\ 0 & \mathcal{T}_0 \end{bmatrix}}^{\mathcal{T}} \begin{bmatrix} \dot{\mathbf{u}}_1 \\ \dot{\mathbf{u}}_2 \end{bmatrix} = \overbrace{\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}}^{\mathcal{A}} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix}.$$

where $\mathbf{u}_1 = \dot{u}$ and $\mathbf{u}_2 = u_{xx}$ and where

$$(\mathcal{T}_0\mathbf{x})(s) := \int_0^1 G_T(s,\theta)\mathbf{x}(\theta)d\theta \quad G_T(s,\theta) = \begin{cases} -\theta & \theta < s \\ -s & s < \theta. \end{cases}$$

To apply the stability conditions of Lemma 3, we observe that $\mathcal{T}_0 = -\mathcal{R}^*\mathcal{R}$ for

$$(\mathcal{R}\mathbf{x})(s) := \int_0^1 G_R(s,\theta)\mathbf{x}(\theta)d\theta \quad G_R(s,\theta) = \begin{cases} 0 & \theta < s \\ -1 & s < \theta. \end{cases}$$

Thus if we choose $\mathcal{Q} = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}$, we find that

$$\mathcal{T}_0^*\mathcal{T}_0 = \mathcal{R}^*\mathcal{R}\mathcal{R}^*\mathcal{R} \leq \|\mathcal{R}\mathcal{R}^*\|\mathcal{R}^*\mathcal{R} = -\|\mathcal{T}_0\|\mathcal{T}_0$$

and

$$\mathcal{Q}\mathcal{T} = \mathcal{T}^*\mathcal{Q}^* = \begin{bmatrix} I & 0 \\ 0 & -\mathcal{T}_0 \end{bmatrix}$$
$$\geq \min\left\{1, \frac{1}{\|\mathcal{T}_0\|}\right\} \begin{bmatrix} I & 0 \\ 0 & \mathcal{T}_0^*\mathcal{T}_0 \end{bmatrix} = \min\left\{1, \frac{1}{\|\mathcal{T}_0\|}\right\} \mathcal{T}^*\mathcal{T}.$$

Furthermore,

$$\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} + \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} = 0.$$

Hence we may use Lemma 7 to conclude PIE to PDE stability of the wave equation.

### B. Operator Inequalities for Exponential Stability

We combine $V_1$ and $V_2$ with Lemmas 4 and 6 to obtain the following operator inequalities for each category of exponential stability.

*Lemma 8 (LPIs for Exp. Stability):* Given $\mathcal{T}, \mathcal{A}$, the PIE $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) Exp. PIE to PDE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon\mathcal{T}^*\mathcal{T}$, $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq -\alpha\mathcal{Q}\mathcal{T}$
2) Exp. PIE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon I$ and $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq -\alpha\mathcal{Q}\mathcal{T}$
3) Exp. PDE stable if $\mathcal{P} \geq \epsilon I$, $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq -\alpha\mathcal{T}^*\mathcal{P}\mathcal{T}$
4) Exp. PDE to PIE stable if $\mathcal{Q}\mathcal{T} = (\mathcal{Q}\mathcal{T})^* \geq \epsilon I$, $\mathcal{Q}\mathcal{T} \leq C\mathcal{T}^*\mathcal{T}$, and $\mathcal{Q}\mathcal{A} + \mathcal{A}^*\mathcal{Q}^* \leq -\alpha\mathcal{Q}\mathcal{T}$

for some $\mathcal{Q}$ or $\mathcal{P}$ and $\epsilon, C, \alpha > 0$.

If we wish to avoid bilinearity in decision variables $\alpha$ and $\mathbb{Q}$ or $\mathcal{P}$, we may replace the Lyapunov negativity conditions in Lem. 4 with PIE or PDE negativity. Recalling the relationship between Lyapunov, PDE, and PIE negativity, we have the following.

*Lemma 9 (Alternative LPIs for Exp. stability):* Given $\mathcal{T}, \mathcal{A}$, the PIE $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) Exp. PIE to PDE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* \geq \epsilon \mathcal{T}^*\mathcal{T}$, $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha I$
2) Exp. PIE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* = \mathbb{Q}\mathcal{T} \geq \epsilon I$ and $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha I$
3) Exp. PDE stable if $\mathcal{P} \geq \epsilon I$, $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq -\alpha \mathcal{T}^*\mathcal{T}$
4) Exp. PDE to PIE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* \geq \epsilon I$, $\mathbb{Q}\mathcal{T} \leq C\mathcal{T}^*\mathcal{T}$, and $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha I$

for some $\mathbb{Q}$ or $\mathcal{P}$ and $\epsilon, C, \alpha > 0$.

*Proof:* Statements 1) and 2) replace Lyapunov negative with PIE negative and bounded. Statements 3) and 4) replace Lyapunov negative with PDE negative and bounded. ■

*Example 2 (The Heat Equation is Exp. PDE stable):* The heat equation $\mathbf{u}_t = \mathbf{u}_{ss} + \lambda \mathbf{u}$ with Dirichlet boundary conditions admits a PIE representation $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x} = (I + \lambda\mathcal{T})\mathbf{x}$ where

$$(\mathcal{T}\mathbf{x})(s) = \int_0^s \theta(s-1)\mathbf{x}(\theta)d\theta + \int_s^1 s(\theta-1)\mathbf{x}(\theta)d\theta$$

and $\mathcal{T} = \mathcal{T}^* = -\mathcal{M}^*\mathcal{M}$ where

$$(\mathcal{M}\mathbf{x})(s) = \int_0^s \theta\mathbf{x}(\theta)d\theta + \int_s^1 (\theta-1)\mathbf{x}(\theta)d\theta.$$

Now, $-\mathcal{M}^*\mathcal{M} \leq -\pi^2\mathcal{T}^*\mathcal{T}$. Hence for $\mathcal{P} = I$, we have that

$$\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} = -2(\mathcal{M}^*\mathcal{M} - \lambda\mathcal{T}^*\mathcal{T}) \leq -2(\pi^2 - \lambda)\mathcal{T}^*\mathcal{T}.$$

Hence by Lemma 8 we have that the heat equation is exponentially PDE stable for any $\lambda < \pi^2$.

### C. Operator Inequalities for Finite Energy Stability

Finally, we combine $V_1$ and $V_2$ with Lemmas 5 and 6 to obtain the following operator inequalities for each category of finite energy stability.

*Lemma 10:* Given $\mathcal{T}, \mathcal{A}$, the PIE $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ is

1) FE PIE to PDE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* \geq 0$, $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha\mathcal{T}^*\mathcal{T}$.
2) FE PIE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* \geq 0$ and $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha I$.
3) FE PDE stable if $\mathcal{P} \geq 0$, $\mathcal{T}^*\mathcal{P}\mathcal{A} + \mathcal{A}^*\mathcal{P}\mathcal{T} \leq -\alpha\mathcal{T}^*\mathcal{T}$.
4) FE PDE to PIE stable if $\mathbb{Q}\mathcal{T} = (\mathbb{Q}\mathcal{T})^* \geq 0$, $\mathbb{Q}\mathcal{T} \leq C\mathcal{T}^*\mathcal{T}$, and $\mathbb{Q}\mathcal{A} + \mathcal{A}^*\mathbb{Q}^* \leq -\alpha I$.

for some $\mathbb{Q}$ or $\mathcal{P}$ and $\epsilon, C, \alpha > 0$.
Note that unlike Lyapunov or exponential PIE stability, FE PIE stability is often possible to verify.

## VI. Summary

Table I summarizes the weakest Lyapunov conditions required for each notion of stability.

## VII. Conclusion

We have proposed a hierarchical classification of Lyapunov, exponential, and finite-energy stability notions in terms of the fundamental state, $\mathbf{x}$, which is related to the PDE state as $\mathbf{x} = D^\alpha\mathbf{u}$ and $\mathbf{u} = \mathcal{T}\mathbf{x}$ and which satisfies $\mathcal{T}\dot{\mathbf{x}} = \mathcal{A}\mathbf{x}$ for some PI operators, $\mathcal{T}, \mathcal{A}$. This framework allows us to unify many existing Lyapunov stability conditions which may or may not include partial derivatives of the PDE state in Lyapunov function lower bounds, upper bounds, and negativity conditions. Specifically, Lyapnunov conditions which establish stability, but not PDE stability will typically imply PIE to PDE stability. This framework allows for classification of PDEs such as the wave and beam equations, which are not PDE stable. Furthermore, the weakest possible Lyapunov and operator $L_2$-inequality conditions are provided for each notion of stability. Such conditions may be verified analytically or using numerical software such as PIETOOLS.

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| | Positivity | | Upper Bound | | Negativity | | |
|---|---|---|---|---|---|---|---|
| Notion | PIE | PDE | PDE | PIE | PIE | PDE | SD |
| PIE2PDE | | X | | X | | | X |
| PIE | X | | | X | | | X |
| PDE | | X | X | | | | X |
| PDE2PIE | X | | X | | | | X |
| Exp PIE2PDE | | X | | X | X | | |
| Exp PIE2PDE | | X | X | | | X | |
| Exp PIE | X | | | X | X | | |
| Exp PIE | X | | X | | | X | |
| Exp PDE | | X | X | | | X | |
| Exp PDE2PIE | X | | X | | | X | |
| FE PIE2PDE | | | | X | | X | |
| FE PIE | | | | X | X | | |
| FE PDE | | | X | | | X | |
| FE PDE2PIE | | | X | | X | | |

TABLE I: The weakest Lyapunov conditions which imply each given notion of stability (2 variants for PIE2PDE and PIE exp. stability). Categories of positivity, bound, and negativity are ordered from strongest to weakest.